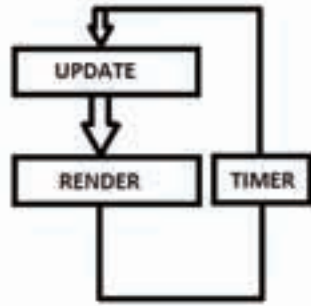


graphics, input, networking and other systems. The library of core functions that are created by a set of programmers are used by other programmers and use these functions to code for other parts of the game. Different types of genres require different types of requirements according to which Game Engines are made.



3rd Loop flow diagram

Technology requirements of different genres

- ▶ **First-Person Shooter Games:** Require efficient rendering, physics-based animation and AI of NPCs.
- ▶ **Platformer and Third-Person Shooter games:** Requires dynamic world, high fidelity animation of main character, camera collision system.
- ▶ **Fighting/Action games:** Requires animation database, accurate user input, character animation.
- ▶ **Racing games:** Requires level of detail, Rendering, Rigid body physics and deformations.
- ▶ **Real-Time Strategy (RTS):** Requires good AI, proper programming for evolving environment.
- ▶ **Massively Multiplayer Online Games (MMOG):** Requires data optimization, memory and account management, VoIP service, intensive use of network.

Game engine architecture

There are three layers: Game Engine API, Hardware Abstraction Layer, and Hardware Layer. Hardware Layer consists of sound card, Graphics card and physics card. In the Game Engine API, engines for specific tasks are made:

Graphics engine

- ▶ Handles the Graphic rendering of the game.
- ▶ Used to model data
- ▶ SDKs like Microsoft DirectX or OGRE can be used to integrate with the game engine.
- ▶ The Graphic Engine is built on the top of human interface library.